

Business Value Mapping

Cloud Engineers cleared positions
What problems are they hiring for?

1. Cloud Modernization Under Security Constraints

- Agencies still rely on outdated, manually managed systems
- They need engineers who can rearchitect workloads into AWS (C2S, GovCloud, IL5-IL6) while preserving data integrity, compliance, and performance.
- Key challenge: balancing agility (cloud innovation) with strict DoD/AIC security compliance (STIGs, RMF, NIST 800-53)

2. Automation & Infrastructure-as-Code at Scale

- Manual provisioning creates inconsistency and risk
- They want IaC experts (Terraform, CloudFormation, Ansible) to standardize deployments, enable rapid rollback, and eliminate drift.

3. Operational Continuity in Classified Environments

- Production support and rapid troubleshooting are mission-critical.
- These environments can't afford downtime; engineers must handle 24/7 operations, ticket triage, and incident response while maintaining classified system uptime.

4. Bridging Development and Operations (DevSecOps)

- Developers ship code; operators manage uptime; the gap causes delays and vulnerabilities
- They're looking for engineers who can deploy, test, and secure code in CI/CD pipelines (Jenkins, GitLab) and ensure containerized environments (Docker / Kubernetes) meet mission requirements.

5. Cloud Security and Compliance Governance

- Each role emphasizes governance, encryption, least privilege, and secure image management.
- Teams must enforce policies across multiple classification levels.

6. Talent Scarcity for Cleared Cloud Engineers

- The clearance + AWS skill combo is rare
- Agencies can't meet transformation goals without cleared talent fluent in both cloud engineering and government compliance frameworks.

Map problems to AWS solutions
(with cost/security trade-offs noted)

1. Cloud Modernization Under Security Constraints

Problem:

Legacy, on-prem or hybrid environment
government workloads need migration to AWS
while meeting IL5-IL6 security requirements
(classified data, zero downtime).

AWS Solutions:

- AWS GovCloud (US-East/West) → Isolated region for sensitive workloads (Fed RAMP High, ITAR, DoD SRG).
- AWS Control Tower + Service Catalog → Baseline multi-account governance and provisioning
- AWS Landing Zone Accelerator (LZA) → Automated compliant baseline (DoD SRG, NIST 800-53)
- AWS PMS / Snowball Edge → Data migration for disconnected or classified environments
- Cloud Endure / Application Migration Service → Lift-and-shift modernization with minimal downtime

2. Automation & IaC at Scale

Problem:

Manual provisioning across multiple envelopes creates drift, risk, and inconsistent security posture.

AWS Solutions:

- AWS CloudFormation → Native IaC for standard provisioning
- Terraform Enterprise / Open Source → Cloud-agnostic IaC with policy as code (Sentinel)
- AWS Systems Manager (SSM) → Patch, config, and automate tasks across fleets
- AWS Config + AWS Service Catalog → Compliance drift detection and auto-remediation

3. Operational Continuity in Classified Environments

Problem:

Cloud services must maintain uptime and resilience despite isolation and limited internet access.

AWS Solutions:

- Amazon CloudWatch / CloudTrail / AWS Config → Full telemetry pipeline
- AWS Backup + EBS Snapshots + Cross-Region Replication → Automated recovery
- Auto Scaling + Elastic Load Balancing (ELB) → high availability
- AWS Systems Manager Incident Manager → Response playbooks for outages

4. DevSecOps Integration

Problem:

Pipeline fragmentation delays secure deployment and patching cycles.

AWS Solutions:

- AWS Code Pipeline + CodeBuild + CodeDeploy → CI/CD automation integrated with IAM and CloudWatch
- AWS ECR + ECS / Fargate or EKS → Container orchestration
- AWS Inspector / Security Hub / GuardDuty → Vulnerability scanning and runtime threat detection
- AWS Secrets Manager / Parameter Store → Centralized secret rotation

5. Cloud Security & Compliance Workloads

Problem:

Classified workloads must remain auditable, encrypted, and least-privileged across all domains.

AWS Solutions:

- AWS Organizations + SCPs → Centralized access and guardrails
- AWS IAM Identity Center (SSO) → Role-based access control
- AWS KMS / Cloud HSM → Data-at-rest encryption and key isolation
- AWS Macie / GuardDuty / Security Hub → Data classification and anomaly detection
- AWS Audit Manager → Automated evidence collection for RAMP/NIST compliance

6. Talent Scarcity & Knowledge Scalability

Problem:

High turnover and skills gap among cleared engineers cause operational bottlenecks

AWS Solutions:

- AWS CloudFormation StackSets + Service Catalog Portfolios → Enable repeatable self-service provisioning
- AWS Systems Manager Documents (Runbooks) → Standardize incident response
- AWS Well-Architected Tool → Institutionalize best practices.